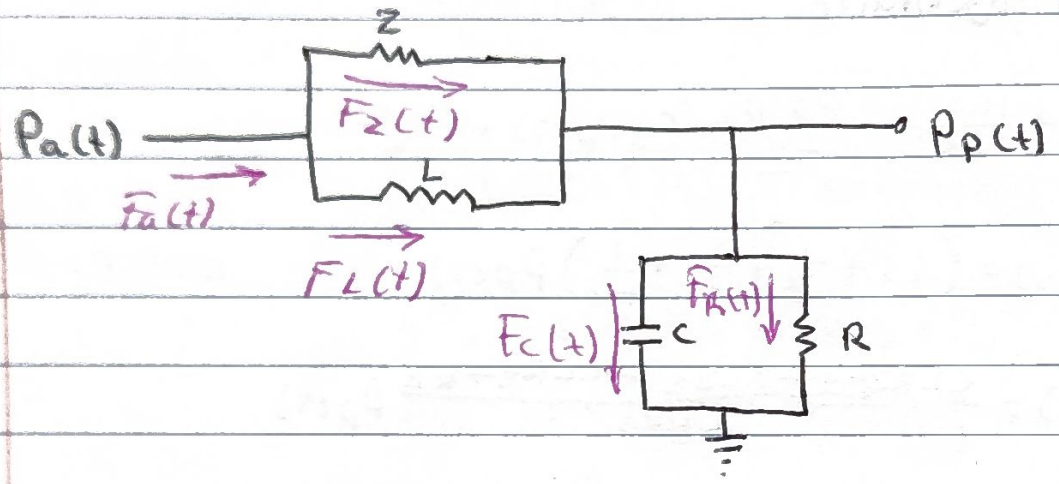
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$L = 100\,mH \quad R = 10\,k \quad C = 470\,mF$



Función de transferencia

$$F_a(t) = F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_Z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

$$F_C(t) = C\,\frac{d\,P_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Transformada de Laplace

$$F_a(s) = F_Z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$F_Z(s) = \frac{P_a(s) - P_p(s)}{Z}$$

$$F_L(s) = \frac{P_a(s) - P_p(s)}{Ls}$$

$$F_C(s) = Cs\,P_p(s)$$

$$F_R(s) = \frac{P_p(s)}{R}$$

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$$\frac{P_a(s)-P_p(s)}{Z}+\frac{P_a(s)-P_p(s)}{Ls}=(Cs\,P_p(s)+\frac{P_p(s)}{R}$$

Procedimiento algebraico

$$\frac{P_a(s)}{Z}-\frac{P_p(s)}{Z}+\frac{P_a(s)}{Ls}-\frac{P_p(s)}{Ls}=(Cs\,P_p(s)+\frac{P_p(s)}{R}$$

$$\left(\frac{1}{Z}+\frac{1}{Ls}\right)P_a(s)=\left(Cs+\frac{1}{R}+\frac{1}{Z}+\frac{1}{Ls}\right)P_p(s)$$

$$\frac{Ls+Z}{LZs}P_a(s)=\frac{CLRZs^2+CLZs+CLRs+RZ}{LRZs}P_p(s)$$

$$\frac{Ls+Z}{LZs}P_a(s)=\frac{CLRZs^2+L(Z+R)s+RZ}{LRZs}P_p(s)$$

$$\frac{P_p(s)}{P_a(s)}=\frac{\frac{Ls+Z}{LZs}}{\frac{CLRZs^2+L(Z+R)s+RZ}{LRZs}}$$

$$\frac{P_p(s)}{P_a(s)}=\frac{LRs+RZ}{CLRZs^2+L(Z+R)s+RZ}$$

Error estacionario

Este error se calcula cuando la entrada [Pa(s)] es un escalón unitario, es decir, $P_a(s)=\frac{1}{s}$:

$$e(s)=\lim_{s\to 0}s\,P_a(s)\left(1-\frac{P_p(s)}{P_a(s)}\right)=\lim_{s\to 0}s\frac{1}{s}\left(1-\frac{LRs+RZ}{CLRZs^2+L(Z+R)s+RZ}\right)=0$$

$e(t)=0V.$

Estabilidad del sistema en lazo abierto

Para analizar la estabilidad del sistema en lazo abierto se deben calcular los polos:

$$CLRZs^2+L(Z+R)s+RZ=0$$

$a=CLRZ \quad b=L(Z+R) \quad c=RZ$

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$$\lambda_{1,2} = \frac{-L(Z+R) \pm \sqrt{L^2(Z+R)^2 - 4CLR^2Z^2}}{2CLRZ}$$

HIPOTENSO

$Z = 0.010$

$L = 0.001$

$R = 1.000$

$C = 3.500$

$$\frac{P_{D(s)}}{P_{a(s)}} = \frac{LRs + Rz}{CLRZs^2 + L(Z+R)s + RZ} = \frac{0.001s + 0.01}{3.5\times10^{-5}s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -14.429 + 8.8051i,$

$\lambda_2 = -14.429 - 8.8051i$

Por lo tanto, el sistema presenta una respuesta estable y subamortiguada.

NORMOTENSO

$Z = 0.010$

$L = 0.001$

$R = 1.000$

$C = 1.500$

$$\frac{P_{D(s)}}{P_{a(s)}} = \frac{LRs + Rz}{CLRZs^2 + L(Z+R)s + RZ} = \frac{0.001s + 0.01}{1.5\times10^{-5}s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -12.062,$

$\lambda_2 = -55.272$

Por lo tanto, el sistema presenta una respuesta estable y sobreamortiguada.

HIPERTENSO

$Z = 0.010$

$L = 0.001$

$R = 1.000$

$C = 1.000$

$$\frac{P_{D(s)}}{P_{a(s)}} = \frac{LRs + Rz}{CLRZs^2 + L(Z+R)s + RZ} = \frac{0.001s + 0.01}{0.00001s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -10.0$

$\lambda_2 = -10.0$

Por lo tanto el sistema presenta una respuesta críticamente amortiguada y estable.

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Modelo de ecuaciones integro-diferenciales

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{P_a(t) - P_p(t)}{Z} + \frac{1}{L}\int[P_a(t) - P_p(t)]dt = C\frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$\frac{P_a(t)}{Z} + \frac{1}{L}\int[P_a(t) - P_p(t)]dt - C\frac{dP_p(t)}{dt} = \frac{P_p(t)}{R} + \frac{P_p(t)}{Z}$$

$$P_p(t)\left(\frac{1}{R} + \frac{1}{Z}\right) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int[P_a(t) - P_p(t)]dt - C\frac{dP_p(t)}{dt}\right]$$

$$P_p(t)\frac{Z+R}{RZ} = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int[P_a(t) - P_p(t)]dt - C\frac{dP_p(t)}{dt}\right]$$

$$P_p(t) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int[P_a(t) - P_p(t)]dt - C\frac{dP_p(t)}{dt}\right]\frac{RZ}{Z+R}$$